

The empirical recurrence for OEIS A232364: recovering a conjecture that is not written down

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Abstract

OEIS A232364 records “Empirical recurrence of order 60”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 157$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 60, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 60 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A232364 is “Number of $n \times 4$ 0..2 arrays with every 0 next to a 1 and every 1 next to a 2 horizontally or vertically.”. It has offset 1 and begins

15, 887, 39321, 1704619, 74665511, 3271362341, 143280358783, 6275473692531, ...

The entry states:

Empirical recurrence of order 60 (see link above)

As of the “Last modified” line on the live entry (May 03 2021 15:06:18, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 6643$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 157$ of the 6643, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 157$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 13286$ exact terms of the model returns order 60 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{60} c_i a(n-i),$$

c_1	38	c_{16}	−1895638712377	c_{31}	962689527139825	c_{46}	−6921211130103790
c_2	188	c_{17}	−6463690706919	c_{32}	3138007593784801	c_{47}	−283086253547660
c_3	2715	c_{18}	9294827476192	c_{33}	−3018729735352888	c_{48}	6137467759538021
c_4	10152	c_{19}	22632871029823	c_{34}	−776055288797029	c_{49}	−4814786458515376
c_5	−88788	c_{20}	−9199365233391	c_{35}	−7461421320816496	c_{50}	2388649635968027
c_6	−773285	c_{21}	53123689935372	c_{36}	−16140461706559689	c_{51}	−370274897578277
c_7	−3735552	c_{22}	−14727626852749	c_{37}	−293236559745317	c_{52}	−734172810129769
c_8	4083582	c_{23}	−238655992102509	c_{38}	−42165068399997838	c_{53}	612756580722936
c_9	185238785	c_{24}	58163088209009	c_{39}	26599218546577645	c_{54}	−262398142553096
c_{10}	334518309	c_{25}	−234298336859605	c_{40}	−67528610088953744	c_{55}	38283926805680
c_{11}	−3340936531	c_{26}	−229961525319125	c_{41}	55445447235957220	c_{56}	42997819778736
c_{12}	−9073447660	c_{27}	690858793677700	c_{42}	−69481482634299059	c_{57}	−25849123637760
c_{13}	18990324942	c_{28}	−286630204102664	c_{43}	54930014551283273	c_{58}	927851040000
c_{14}	170964511768	c_{29}	1509415818583987	c_{44}	−40770455953347066	c_{59}	2402597376000
c_{15}	387794605737	c_{30}	1307195782016285	c_{45}	25727417545193390	c_{60}	−444165120000

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 60, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $60 + 4$ terms removed returns order 60 as well, so $\deg q_{\min} = 60 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $60 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 60 that this sequence satisfies — and that family turns out to have exactly one member.

References

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